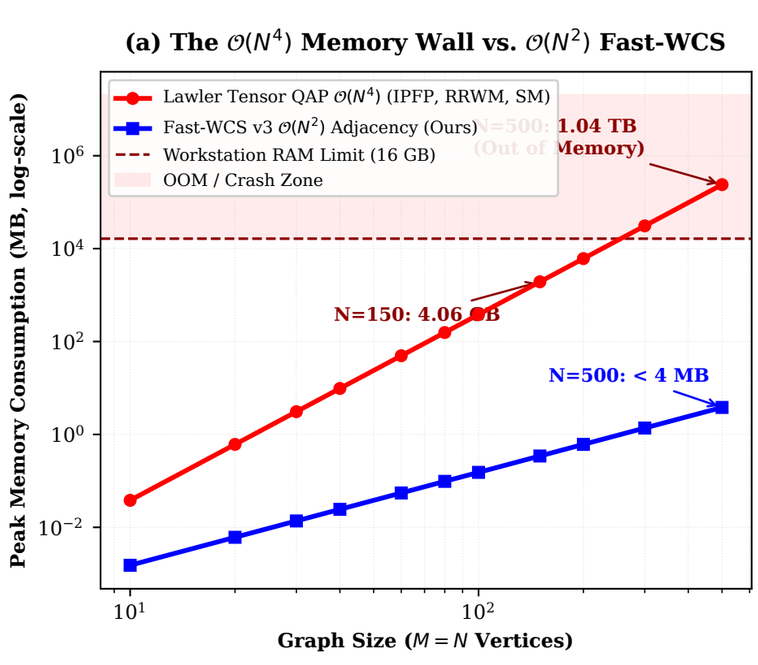
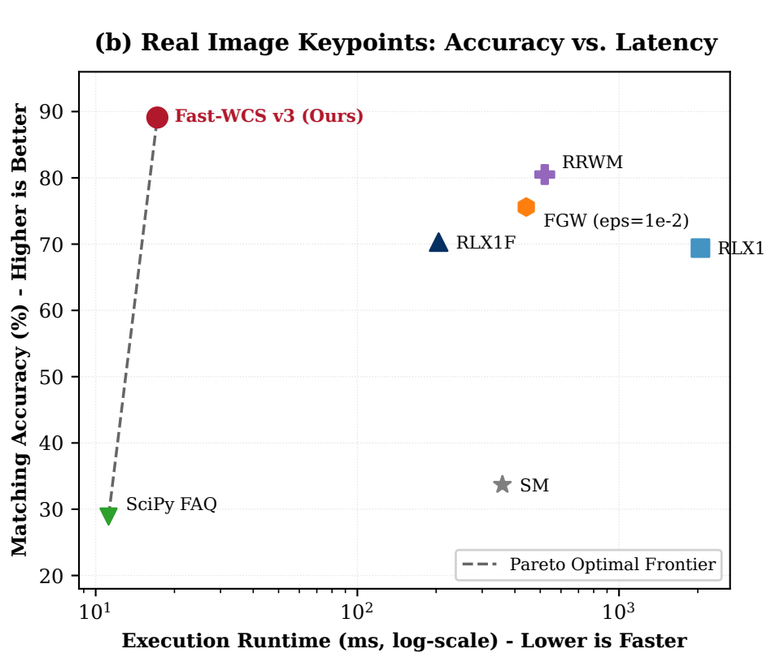
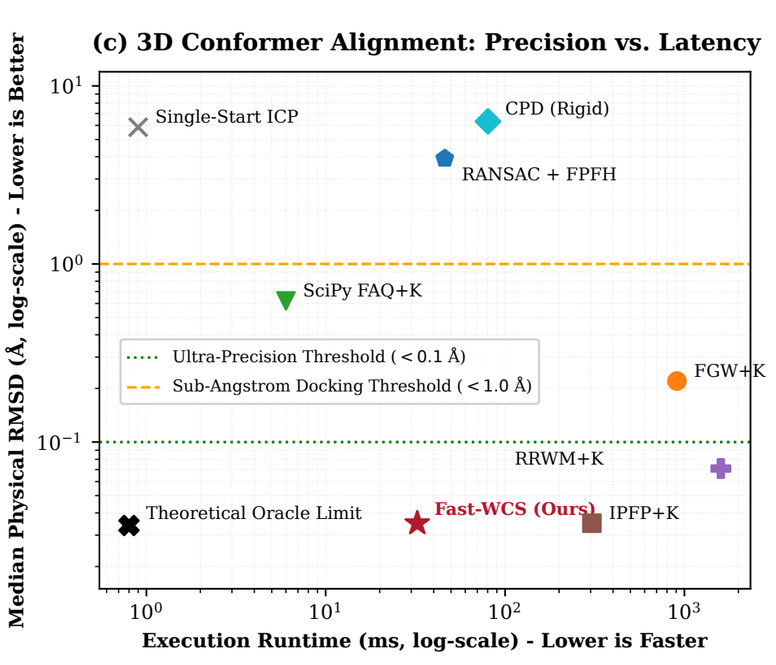

(a) The $\mathcal{O}(N^4)$ Memory Wall vs. $\mathcal{O}(N^2)$ Fast-WCS
Lawler Tensor QAP $\mathcal{O}(N^4)$ (IPFP, RRWM, SM)
Fast-WCS v3 $\mathcal{O}(N^2)$ Adjacency (Ours)
Workstation RAM Limit (16 GB)
OOM / Crash Zone
N=500: 1.04 TB
(Out of Memory)
N=150: 4.06 GB
N=500: < 4 MB
Peak Memory Consumption (MB, log-scale)
Graph Size ($M = N$ Vertices)


(b) Real Image Keypoints: Accuracy vs. Latency
Fast-WCS v3 (Ours)
RRWM
FGW (eps=1e-2)
RLX1F
RLX1
SciPy FAQ
SM
Pareto Optimal Frontier
Matching Accuracy (%) - Higher is Better
Execution Runtime (ms, log-scale) - Lower is Faster


(c) 3D Conformer Alignment: Precision vs. Latency
Single-Start ICP
CPD (Rigid)
RANSAC + FPFH
SciPy FAQ+K
Ultra-Precision Threshold (< 0.1 Å)
Sub-Angstrom Docking Threshold (< 1.0 Å)
FGW+K
RRWM+K
Theoretical Oracle Limit
Fast-WCS (Ours)
IPFP+K
Median Physical RMSD (Å, log-scale) - Lower is Better
Execution Runtime (ms, log-scale) - Lower is Faster